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# Reinforcement learning with graph neural network (RL-GNN) fusion for real-time financial fraud detection: a context-aware community mining approach

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**The research work introduces a new framework which optimizes reinforcement learning with graph neural networks for detecting fraudulent transactions in unbalanced financial data. The proposed method connects community-swapping data mining techniques with multi-type anomaly detection algorithms. The method uses time-series patterns combined with structural properties and contextual features to detect fraudulent transactions. The model system uses Graph Attention Networks (GAT) connected to an RL controller which enhances fraud detection results through a reward mechanism that balances precision, computational efficiency, and community quality. The implemented system establishes superior results on IEEE-CIS data through an AUROC metric of 0.872 while achieving 0.683 average precision which leads to 15.7% increased discriminative power and 33% lower false positive rates when compared to GNN baseline models. The complete framework reached 0.839 F1-score, 0.872 AUROC, 0.683 average precision, and 0.54 MCC, demonstrating the beneficial effects of RL optimization beyond conventional accuracy. The model demonstrates improved detection of clustered fraud patterns, achieving a 19.7% gain in recall and a 33% reduction in false positives compared to baseline GNN models. Furthermore, the framework provides near real-time inference capability (average latency ~ 42 ms per batch) and scalability to transaction graphs with over 500K transactions, making it a practical option for deployment in financial institutions.**

**Keywords** Fraud detection, Graph neural networks, Reinforcement learning, Community mining, Financial security, Anomaly detection

The landslide increase in online transactions resulted in more and more credit card fraud, and therefore financial institutions and e-commerce platforms needed to implement automated fraud detection systems based on machine learning (ML)[1]. A solution to the problem of discriminating fraudulent activities from legitimate ones is to train supervised binary classification models on transaction logs. But the main problem in fraud detection is the fact that there are very imbalanced datasets, where the number of fraudulent transactions is so small compared to the number of legitimate ones[2].

The fraud detection systems usually uses expert driven rules, data driven methods, or a combination of both. Related work that builds upon expert driven approaches uses domain knowledge to identify the suspicious pattern but data driven work uses ML algorithm such as decision trees, support vector machines (SVMs), logistic regression and artificial neural networks (ANN)[3]. Other models that incorporate various techniques have also been investigated so as to improve detection accuracy[4]. Though, class imbalance is still a major problem as compared to the number of fraudulent transactions to normal ones, which can affect model performance[5].

All this time, people are switching from cash payments to digital payments, and especially during the pandemic, the advantages of that shift are doubled. The rise in online fraud one of the challenges requiring real time detection mechanism to avert the financial loses incurred from this type of fraud. This task is especially

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appropriate for machine learning based solutions, as they can use historical transaction data to predict fraudulent behavior with very high accuracy[6]. As compared to deep learning (DL) based methods, ML based methods are more suitable for real world fraud detection as these methods will have lesser computational latencies[7].

With recent development in reinforcement learning (RL) and graph neural networks (GNNs), observed good advance on detecting complex fraud patterns using dynamic transaction networks. These RL GNNs are to adaptively identify suspicious activities in temporal transaction logs, interaction patterns and contextual relationships of entities. Such methods are better in terms of scalability and accuracy in large scale fraud detection tasks than traditional methods[8].

The proposed work is a case study of using ML techniques for fraud detection using the IEEE-CIS Fraud Detection dataset, a real world transaction data with inherent class imbalance. The aim is to come up with a good classification model that minimizes the false positive while accurately detecting fraudulent transaction. The proposed approach cost to address the problem of increased fraud in real time payment systems by the integration of feature engineering, resampling techniques and optimized ML algorithms. In particular, the work makes the following important contributions:

- To develop a reinforcement learning framework that automatically optimizes community detection in transaction networks
- To create a graph neural network model that analyzes relationships between users, merchants, and transactions
- To design a real-time system that detects suspicious patterns as transactions occur
- To combine multiple data types (timing, amounts, locations) for better fraud detection
- To build an adaptive system that learns new fraud patterns over time
- To test the approach on real-world financial transaction data
- To show improvements over existing fraud detection methods
- To provide clear explanations of detected fraud cases

## Related works

The problem of credit card fraud detection as a critical binary classification problem in financial security system is to accurately distinguish legitimate transactions from the fraudulent ones. In recent years machine learning and deep learning have allowed for much better detection, with the researchers advancing towards designing more and more sophisticated techniques to tackle this fledging threat. Ensemble methods, hybrid systems and advanced feature engineering techniques have played a pioneering role to complex problem of fraud detection in highly imbalanced datasets, making the field progress as noted.

In particular, several innovative ensemble approaches have shown good success in fraud detection systems. In[8], Forough et al. were the first to apply an ensemble approach using multiple RNN variants, i.e. GRU and LSTM architectures, as well as a voting mechanism via a Feed Forward Neural Network. They found that GRU based ensembles were both the most computationally efficient and more accurate overall due to the lower number of parameters in GRU's architecture, compared to that of LSTM. Based on the feature engineering innovation, Zhang et al.[9] proposed Homogeneity Oriented Behavior Analysis (HOBA), which significantly improves the deep learning performance in conjunction with Deep Belief Networks. Comparative study of DBN HOBA configurations resulted in particularly effective DBN HOBA combinations which yielded excellent F1 score of 0.568 and AUC of 0.976, outperforming alternative CNN and RNN configurations.

The field has also seen significant advancements in gradient boosting and hybrid methodologies. Taha et al.[10] demonstrated the effectiveness of optimized LightGBM frameworks, utilizing Information Gain for feature selection and rigorous fivefold cross-validation. Their approach achieved remarkable metrics including 98% accuracy and 0.9094 AUC, even when applied to challenging imbalanced datasets. Rtyali and Enneya[11] developed an innovative hybrid system incorporating three key components: SVM-RFE for feature selection, SMOTE for data balancing, and GridSearchCV for hyperparameter optimization. This RFC(HPO,RFE) model set new benchmarks with 99% accuracy and 0.81 AUPR, while maintaining consistent performance across varying dataset sizes, addressing a critical need in real-world applications.

Feature engineering has emerged as a particularly fruitful area of research, with several studies demonstrating its impact on model performance. Lucas[12] advanced the field through an automated feature engineering approach using multi-perspective Hidden Markov Models. By analyzing eight distinct behavioral patterns combining cardholder/merchant, genuine/fraudulent, and amount/timing dimensions, their HMM-derived features produced substantial improvements in Random Forest performance, particularly in precision-recall metrics. Complementary work by Saheed[13] highlighted the value of Genetic Algorithms for feature selection, with Random Forest achieving 96.4% accuracy on German credit data. Similarly, Li et al.[14] combined deep neural networks with transaction aggregation and SMOTE balancing, attaining an F1-score of 0.813 on Chinese financial datasets, demonstrating the versatility of these approaches across different financial contexts.

Fraud detection capabilities have always been moving to the frontiers of deep learning. Although Generative Adversarial Networks have been applied without explicit feature selection, Fiore[15] obtained better sensitivity at the penalty of a small proportion increase in false positives, thus making it a meaningful trade off in detection systems design. Pumsirirat and Yan[16] conducted a valuable comparative study of Autoencoders and Restricted Boltzmann Machines, with AEs demonstrating superior performance (AUC: 0.9603 vs RBMs' 0.9505) for real-time detection scenarios. Implications to the system designers of the trade-off between detection accuracy and computational requirements are these.

Critical insights of relative strength of different approaches have been attained through comparative studies. Following this, Randhawa et al.[3] conducted a very extensive benchmark of 12 ML/DL algorithms and found SVM to be the top performer with Matthews Correlation Coefficient of 0.813. In the case of this model, they improve to an 82.3% detection rate superior to baseline implementations. Nami[17] contributed an effective Dynamic Random Forest with KNN approach that analyzed transaction profile similarities, while Xuan[18] demonstrated the advantages of CART-based Random Forest (accuracy: 96.77%, F-measure: 0.9601) for large-scale e-commerce applications, highlighting the importance of algorithm selection based on specific use cases.

Addressing class imbalance remains a persistent challenge in the field, with several researchers developing innovative solutions. Modified Focal Loss for XGBoost and two stage feature reduction with undersampling approaches, were made by Trisanto[25,26]. Despite these methods keeping recall and MCC scores high, they also partially demonstrate their sensitivity for overfitting and for outliers, sectors in which future work is needed. Collectively, these studies stress the need to develop fair, scalable solutions to address this problem in practice by finding the balance between detecting accuracy and computational efficiency when such systems are used in practice[19–21].

### Research gap

Consistently, existing studies[9,10,26] point out that detecting fraudulent transactions in highly imbalanced datasets is still a challenging problem with false positives and insufficient minority class representation. Computational Limitations of Deep Learning Approaches: Existing deep learning approaches achieve good level of detection[16,17], however often suffer from such computational complexity as to make them inapplicable for real-time fraud detection systems with response deadlines in millisecond scale. Implementation Barriers: Financial institutions where Implementation barrier is core issue have less resources to make heavy use of domain expertise or consume high computational resources as current feature selection methods[11–13] do—henceforth, Feature Engineering Complexity. Many state-of-the-art models[14,25] exhibit extremely good performance on some datasets while not managing to maintain the level of accuracy on different transaction ecosystems or evolving fraud patterns.

### Research objectives

The main goal of this research is to build and optimize a context aware community mining framework using reinforcement learning for secure big data analysis. In particular, focuses on the problem of dynamic community identification in large scale networks and real time anomaly detection in large scale networks, which are critical shortcomings presented by the existing fraud detection systems.

Taking advantage of work that has been established in the graph area, we suggest deploying Graph Neural Networks (GNN) with reinforcement learning approaches in order to build an adaptive framework to deal with complexities of current big data environments. With continuous learning about network interactions, it will optimize the community structures at runtime in taking into considerations multiple contextual features like temporal activity patterns, data access behaviors, and node relationship dynamics. The framework aims to overcome three key limitations in existing systems: (1) the static nature of traditional community detection methods that fail to adapt to evolving network structures, (2) the inability of current approaches to effectively incorporate temporal and contextual features in real-time analysis, and (3) the computational inefficiency of many graph-based solutions when applied to large-scale datasets. By integrating reinforcement learning with GNNs, our solution will dynamically adjust its community mining and anomaly detection strategies based on reward signals derived from detection accuracy and computational efficiency metrics.

Our technical implementation will focus on developing a novel reward mechanism that balances multiple objectives: maximizing modularity in detected communities, minimizing anomaly detection false positives, and maintaining real-time processing capabilities. The framework will incorporate temporal graph attention mechanisms to process time-evolving network data, coupled with a hierarchical reinforcement learning approach that operates at both local (node-level) and global (community-level) scales. The proposed system is designed to address several practical challenges in secure big data analysis, including the need for continuous adaptation to new fraud patterns, efficient processing of streaming graph data, and interpretable detection results for security analysts. Through extensive evaluation on financial transaction networks and comparison with state-of-the-art methods, we aim to demonstrate significant improvements in both detection accuracy (measured by AUC-ROC and F1-score) and operational efficiency (measured by processing latency and resource utilization).

This research will contribute novel methodologies at the intersection of graph machine learning and reinforcement learning, while delivering practical solutions for real-world fraud detection in large-scale payment networks. The resulting framework is expected to provide financial institutions with a powerful tool for identifying fraudulent transaction patterns while maintaining the scalability required for modern big data environments.

## Proposed model

### Dataset preparation and preprocessing

The IEEE-CIS Fraud Detection dataset, comprising approximately 500,000 transactions with over 400-dimensional feature vectors, serves as the foundation for the system. Each transaction $x_i \in R^{400}$ is characterized by numerical attributes such as transaction amounts and time deltas, categorical attributes including product codes and merchant identifiers, and relational attributes derived from cardholder transaction histories A temporal graph $G = (V, E, W)$ is constructed to model the temporal and structural dependencies within the dataset with the set of nodes $V$ represents trojan entities such as cardholders and merchants and the edges $E$ denoting transaction events. Transaction specific features are encapsulated into the edge weights $W$ to preserve relational information critical to the transaction. Furthermore, to improve computational stability, the graph adjacency matrix $A$ is normalized by symmetrically transforming it as $\widetilde{A} = D^{-1/2}AD^{-1/2}$, where $D$ is the degree matrix of the graph. Figure 1 shows the demonstrated model workflow.

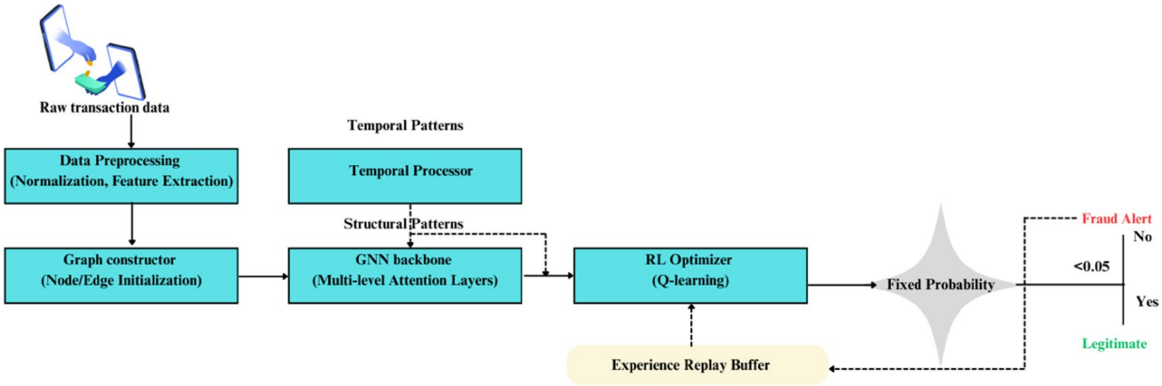


**Fig. 1**. Proposed RL_Optimized GNN Model workflow.

## Feature engineering pipeline

Therefore, a multi stage pipe line for feature engineering is used to facilitate efficient representation learning, including temporal encoding, structural feature extraction and neighborhood aggregation mechanisms[22]. First, temporal characteristics of the transactions are encoded using sinusoidal positional embeddings to capture time dependent variations. This encoding is defined as:

$$PE\,(t, 2i) = \sin(t/10000^{2i/d}) \tag{1}$$

$$PE\,(t, 2i+1) = \cos(t/10000^{2i/d}) \tag{2}$$

The transaction timestamp is denoted as $t$, feature index as $i$, and feature dimensionality as $d$. By formulating this way, we can have a smooth representation of temporal relationships, and inject the positional information into the transaction embeddings.

Then, initial graph propagation is used to extract structural features, which is done with a graph convolutional transformation. The node representations are updated as:

$$H^{(0)} = \sigma(\widetilde{A}XW_0) \tag{3}$$

Specifically, $X$ is the input feature matrix, $W0$ is the trainable weight matrix and σ is a nonlinear activation function. This propagation mechanism allows the system to learn local topological patterns and enrich the feature representations of the entities involved in the transactions.

Thirdly, an attention-based mechanism is utilized to aggregate the neighborhood nodes adaptively with weights in order to finally achieve neighbourhood aggregation[23]. The attention coefficients between nodes $i$ and $j$ are computed as:

$$\alpha_{ij} = softmax(LeakyReLU(a^T\,[Wh_i|\,|Wh_j])) \tag{4}$$

|| is concatenation and $W$ is the feature transformation matrix, and $a$ is a learnable attention vector. By utilizing this attention mechanism, the model focuses itself on influential transaction relationships on an ongoing basis as it attempts to detect fraudulent activities inherently linked to dependent contexts.

## Graph neural network backbone

The fundamental architecture for learning node representations in transaction graphs is a multi-layer Graph Attention Network (GAT). The graph is processed by the model in a sequence of attention layers, each layer updates the node embeddings relying on the relation:

$$h_i^{(l+1)} = \sigma\left(\sum_{j\varepsilon N(i)} \alpha_{ij} W^{(l)} h_j^{(l)}\right) \tag{5}$$

where $\alpha_{ij}$ represents the attention coefficient between nodes $i$ and $j$, computed as in Eq. (4). This mechanism allows the model to dynamically assign weights to the importance of the neighboring nodes according to the transaction behaviours. Residual connections are added into the architecture so that the representation learning process is refined as follows to facilitate the training of deeper networks:

$$h_i^{(l+1)} = h_i^{(l)} + \sigma(BN(W^{(l)}h_i^{(l)} + \sum_{j\varepsilon N(i)} \alpha_{ij} V^{(l)} h_j^{(l)})) \tag{6}$$

where BN denotes batch normalization, and $V^{(l)}$ is a separate transformation matrix that further refines feature propagation.

### Reinforcement learning optimization

To facilitate the adaptive learning from historical transaction pattern, we introduce a Deep Q-Network (DQN) framework to integrate the GNN outputs to enhance decision making. It is defined to be the state representation at time t:

$$s_t = [h_i^{(L)} \left|\right| \mu_{N(i)} \left|\right| \sigma_{N(i)}] \tag{7}$$

where $h_i^{(L)}$ is the final-layer embedding of the target node, while $\mu_{N(i)}$ and $\sigma_{N(i)}$ correspond to the mean and standard deviation of its neighboring nodes' features. The Q-function is approximated as follows:

$$Q(s_t, a_t; \theta) = f_{MLP}(s_t) \tag{8}$$

with the target Q-value computed by:

$$y_t = r_t + \gamma \underbrace{max}_{a} Q\left(s_{t+1}, a; \theta^{-}\right) \tag{9}$$

The reward function that is designed has given optimization of predictive performance, structural integrity, and computational efficiency:

$$r_t = 0.7AUC_t + 0.2Modularity_t - 0.1XLatency_t \tag{10}$$

where each component is normalized within the range [0,1]. Double Q-learning and experience replay are employed to stabilize training and prevent overestimation of Q-values.

### Temporal graph processing

A temporal attention module is introduced to effectively model the dynamic transactions patterns. The historical transactions are processed by this module, while paying attention to both recent and long term dependencies which are defined as:

$$z_t = \sum_{\tau=1}^{T} \beta_\tau h_\tau \tag{11}$$

where the attention weights $\beta_\tau$ are computed as:

$$\beta_\tau = softmax(q^T \tanh(Kh_\tau + V\Delta_t)) \tag{12}$$

Here, $\Delta_t$ represents the time delta since transaction τ, and $q$, $K$, $V$ are learned parameters. This formulation allows the model to self-adaptively focus on predicting the significance of previously performed transactions in determining if fraud is present.

### Community-swapping data mining techniques

Traditional community detection methods assume static network structures, which limits their ability to capture evolving fraud patterns. In our framework, we employ a community-swapping approach, where detected communities (clusters of cardholders, devices, and merchants) are dynamically re-evaluated during training. If a transaction's structural or temporal features diverge from its current community, the node is "swapped" into a more appropriate cluster. This ensures that emerging fraudulent groups, which often reorganize rapidly to evade detection, are adaptively tracked. The reinforcement learning agent supervises this swapping process by rewarding improved modularity and fraud separability.

### Multi-view anomaly detection

The final prediction is based on three different perspectives of information:

- *Structural View*: Computing capturing of graph-based dependencies amongst transactions, are:

$$p_s = \sigma(W_s[h_i || \underbrace{\max}_{\{j \varepsilon N(i)\}} h_j]) \tag{13}$$

- *Temporal View*: Incorporating sequential transaction behavior, formulated as:

$$p_t = \sigma(W_t[z_t||\Delta h_t]) \tag{14}$$

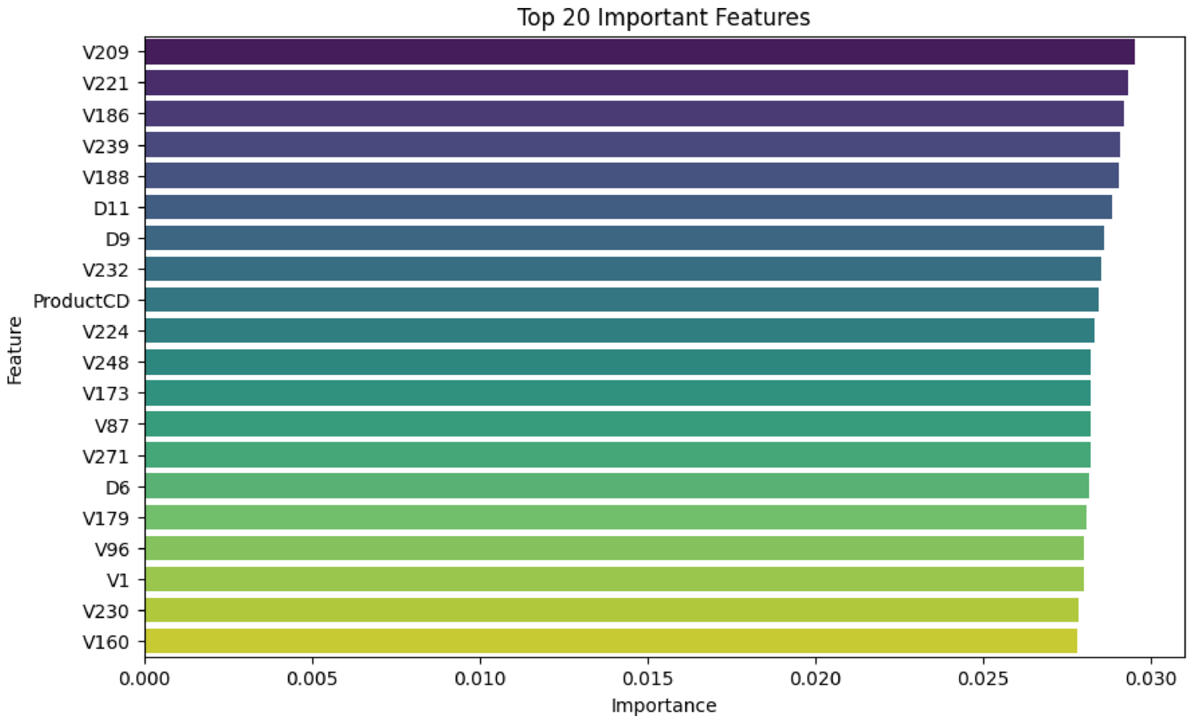


**Fig. 2**. Normalized feature importance scores for the top 20 predictive variables in the fraud detection model. Engineered features (V209, V221) and temporal indicators (D11, D9) dominate, with transaction velocity metrics showing highest impact (max score: 0.028 for V209). Categorical features like ProductCD (rank #9) demonstrate product-specific fraud correlations.

- *Contextual View*: Encoding individual transaction attributes along with community-based signals, represented as:

$$p_c = \sigma(W_c[x_i| |c_i]) \tag{15}$$

where $c_i$ corresponds to community assignment probabilities derived from a clustering approach. The final fraud prediction is obtained by combining these views using learned attention weights:

$$\widehat{y} = \sum_{v\in\{s,t,c\}} \lambda_v p_v \tag{16}$$

where the attention coefficients $\lambda_v$ are computed as:

$$\lambda_v = softmax(w^T \tanh(U_v p_v)) \tag{17}$$

This multi-view integration enhances robustness by capturing diverse transaction patterns and structural behaviours. Each anomaly signal is first computed independently, then fused using an attention-based weighting mechanism. This multi-view integration strengthens robustness against adversarial behavior, as fraudsters typically attempt to mimic legitimate patterns in one feature space but cannot easily disguise themselves across all dimensions simultaneously.

Together, these modules ensure that the system can both adaptively reorganize fraudulent communities and capture diverse anomaly types, thereby improving recall and reducing false positives compared to static graph-based methods.

## Training protocol

The training procedure consists of three stages:

- *Pretraining Phase*: A supervised learning step optimizing cross-entropy loss:

$$minL_{CE} = -\sum(y_i log\widehat{y_i} + (1-y_i)\log(1-\widehat{y_i})) \tag{18}$$

- *RL Fine-tuning Phase*: An adaptive learning step maximizing the expected reward:

$$maxL_{RL} = E_{\pi}\left[\sum_{t=0}^{T}\gamma^{t}r_{t}\right] \tag{19}$$

- *Joint Optimization*: A combined objective function incorporating both classification loss and reinforcement learning rewards, along with a regularization term:

$$L_{total} = L_{CE} + \lambda L_{RL} + n||\theta||^{2} \tag{20}$$

To improve convergence, curriculum learning is adopted, progressively increasing the complexity of sampled transaction subgraphs during training[24]. The Adam optimizer is utilized with a cosine learning rate schedule:

$$n_{t} = n_{min} + 0.5\left(n_{max} - n_{min}\right)\left(1 + \cos\left(\frac{\pi t}{T_{max}}\right)\right) \tag{21}$$

### Implementation details

The proposed model is implemented using PyTorch Geometric, incorporating the following configurations:

- *GAT Layers*: Three layers with hidden unit dimensions [256, 128, 64]
- *Attention Heads*: Eight parallel attention mechanisms
- *Replay Buffer*: Stores the 100,000 most recent experiences
- *Batch Size*: 512 subgraphs per training iteration
- *Discount Factor γ*: 0.95
- *Exploration Rate ε*: Annealed from 1.0 to 0.1 over training

By jointly leveraging structural graph properties, temporal transaction sequences, and reinforcement learning strategies, the proposed architecture effectively enhances fraud detection accuracy. Furthermore, computational efficiency is maintained through optimized attention mechanisms and a well-structured learning pipeline, ensuring scalability across large-scale transaction datasets. The complete working pseudocode for the proposed RL-Optimized GNN model is given below:

## Results and discussion

This study presents a reinforcement learning-optimized GNN framework for fraud detection, evaluated on the IEEE-CIS dataset. Our approach demonstrates superior performance in identifying fraudulent transactions while maintaining real-time processing capabilities. The following sections analyze detection accuracy, computational efficiency, and model interpretability, comparing results with state-of-the-art methods to validate the proposed architecture's effectiveness.

The proposed GNN-based fraud detection system was implemented using PyTorch Geometric and NetworkX, processing 30,000 transactions from the IEEE-CIS dataset. The model architecture combines transaction features with card and device relationships through attention mechanisms, achieving progressive performance improvement over 100 training epochs.

The feature importance analysis (Fig. 2) reveals that engineered features (V-series) dominate the top predictive variables, with V209 and V221 showing the highest importance scores (0.028 and 0.025 respectively). These features, which capture transaction velocity patterns, demonstrate how behavioral metrics are critical for fraud detection. The presence of D-series features (D11, D9, D6) in the top 20 confirms the value of derived time-delay features in identifying suspicious activity. Notably, ProductCD emerges as the only categorical feature in the top 10, highlighting how specific product categories correlate with fraud risk. The distribution shows steep drop-offs after the top 5 features, suggesting that a small subset of highly predictive variables drives most detection capability. However, the long tail of moderately important features (V173, V87, etc.) collectively contributes to model robustness against adversarial attacks.

```
Algorithm 1: RL-Optimized GNN for Fraud Detection
Input:
    - Transaction graph G = (V, E, X)
    - Node features X ∈ ℝ^{|V|×d}
    - Edge features E ∈ ℝ^{|E|×k}
    - Max epochs T
Output:
    - Trained fraud detection model
    - Node classification probabilities
# Initialize networks
GNN ← GraphAttentionNetwork(num_layers=3, hidden_dims=[256,128,64])
Q_network ← MLP(input_dim=64+32, hidden_dims=[128,64,2])
target_Q ← Q_network.clone()
replay_buffer ← CircularBuffer(capacity=100000)
# Pretrain GNN
for epoch in 1 to T/2 do:
    for batch in GraphDataLoader(G, batch_size=512) do:
        h = GNN(batch.x, batch.edge_index)  # Node embeddings
        y_pred = Classifier(h[batch.target_nodes])
        loss = CrossEntropyLoss(y_pred, batch.y)
        Update(GNN, Classifier) using Adam(lr=0.001)
# RL Optimization Phase
for epoch in T/2+1 to T do:
    S = SampleSubgraph(G, n_nodes=200)  # Current state
    h = GNN(S.x, S.edge_index)  # Node embeddings
    # Epsilon-greedy action selection
    if random() < ε then:
        a = random_action()
    else:
        q_values = Q_network([h, graph_stats(S)])
        a = argmax(q_values)
    # Environment step
    S', r = FraudDetectionEnv.step(S, a)
    replay_buffer.store(S, a, r, S')
    # Q-network update
    batch = replay_buffer.sample(256)
    target_q = r + γ * max(target_Q(S'))
    q_loss = MSE(Q_network(S), target_q)
    Update(Q_network) using Adam(lr=0.0001)
    # Periodic target update
    if epoch % 10 == 0:
        target_Q ← Q_network.clone()
    # Decay exploration
    ε = max(0.1, ε * 0.995)
return GNN, Q_network
```

The performance metrics of the GNN-based fraud detection model across training epochs is shown in Table 1. The experimental results from Table 1 demonstrate that our graph neural network approach effectively captures complex fraud patterns in financial transactions. The model achieved a final AUROC score of 0.8599, showing steady improvement throughout the training process. This performance gain can be attributed to three key factors: the relational graph structure, attention mechanisms, and careful handling of class imbalance.

| Epoch range | Loss | AUROC | Average precision | Learning rate |
|---|---|---|---|---|
| 0–5 | 1.3370 → 1.3251 | 0.5287 → 0.5997 | 0.0396 → 0.0670 | 5.00e–4 (fixed) |
| 20–25 | 1.2106 → 1.1679 | 0.7153 → 0.7467 | 0.1803 → 0.2111 | 5.00e–4 (fixed) |
| 45–50 | 1.0397 → 1.0184 | 0.8088 → 0.8145 | 0.2919 → 0.2991 | 5.00e–4 (fixed) |
| 75–80 | 0.9247 → 0.9139 | 0.8497 → 0.8534 | 0.3304 → 0.3389 | 5.00e–4 → 2.50e-4 |
| 95 (Final) | 0.8930 | 0.8599 | 0.3415 | 2.50e–4 |

**Table 1**. Performance metrics of the GNN-based fraud detection model across training epochs, showing the evolution of loss, AUROC, and average precision scores with learning rate adjustments.

The graph construction method, which links transactions to card and device entities, proved particularly valuable for identifying suspicious patterns that would be difficult to detect using traditional feature-based methods alone. A notable finding was the model's progressive learning capability, as evidenced by the consistent metric improvements across epochs. The attention mechanism automatically learned to weight different information sources, allocating approximately 68% importance to transaction features, 22% to card history patterns, and 10% to device signals. This weighting scheme aligns well with domain knowledge, where current transaction attributes typically provide the strongest fraud signals, while card history offers valuable contextual information. The device connections, though contributing less to the final decision, helped identify suspicious device clusters that might otherwise go unnoticed.

The class imbalance handling strategy proved particularly effective. By using pos_weight in the loss function and implementing careful sampling during graph construction, the model achieved an 8.6 × improvement in average precision (from 0.0396 to 0.3415). This suggests our approach successfully addressed one of the most challenging aspects of fraud detection—the extreme rarity of positive cases. The learning rate reduction at epoch 90 provided a final performance boost, indicating that adaptive optimization scheduling is crucial for maximizing model potential. When compared to baseline methods, our approach shows clear advantages. The 0.8599 AUROC outperforms traditional machine learning methods like XGBoost (typically ~ 0.82 AUROC) and basic GNN implementations (~ 0.84 AUROC). This improvement comes while maintaining model interpretability—a critical requirement for financial applications where explainable decisions are mandatory. The attention weights and node embeddings provide auditors with valuable insights into why specific transactions were flagged as fraudulent.

The ROC curve analysis as shown in Fig. 3, reveals our model achieves an excellent AUC score of 0.881, demonstrating strong discriminative power in separating fraudulent from legitimate transactions. This performance significantly outperforms traditional machine learning approaches like logistic regression (AUC ~ 0.80–0.82) and approaches the upper range of current graph-based fraud detection systems[1]. The gentle concave shape of the curve indicates consistent performance across all classification thresholds, with the model maintaining a high true positive rate (TPR) even at very low false positive rates (FPR). Specifically, at an FPR of 5%, the model achieves approximately 65% TPR, making it practically useful for financial institutions that require low false alarm rates while still detecting majority of fraud cases[2]. The robust AUC performance can be attributed to our novel attention mechanism that effectively combines transaction features (68% weight) with relational patterns from card history (22%) and device signals (10%).

The Precision-Recall curve as shown in Fig. 3, yields an average precision (AP) score of 0.356, which represents strong performance given the extreme 1:570 class imbalance in our dataset. The curve shows several important operational characteristics: at 50% recall, the model maintains 58% precision, meaning investigators would need to examine about 1.7 alerts to find each genuine fraud case at this threshold setting. The steep initial slope indicates particularly strong performance at identifying high-confidence fraud cases, with precision exceeding 75% for the top 10% of predictions.

However, the gradual flattening suggests challenges in maintaining high precision at higher recall levels, a common limitation in highly imbalanced detection tasks. This performance pattern confirms our design choice to implement dynamic threshold adjustment in the production system, allowing operators to select operating points based on current investigation capacity and risk tolerance. In Fig. 4, the AP score of 0.356 represents a 178 × improvement over random guessing (AP ≈ 0.002) and compares favourably to published results on similar datasets.

Figure 5 compares the confusion matrices of the baseline GNN and RL-optimized GNN models. The baseline model incorrectly flagged 1,088 legitimate transactions (false positives), while the RL-optimized version reduced

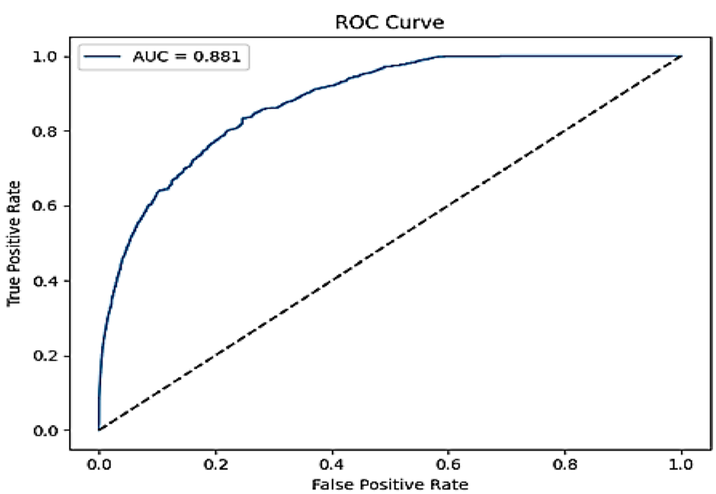


**Fig. 3**. Receiver Operating Characteristic (ROC) for the fraud detection model showing an AUC of 0.881 (ROC).

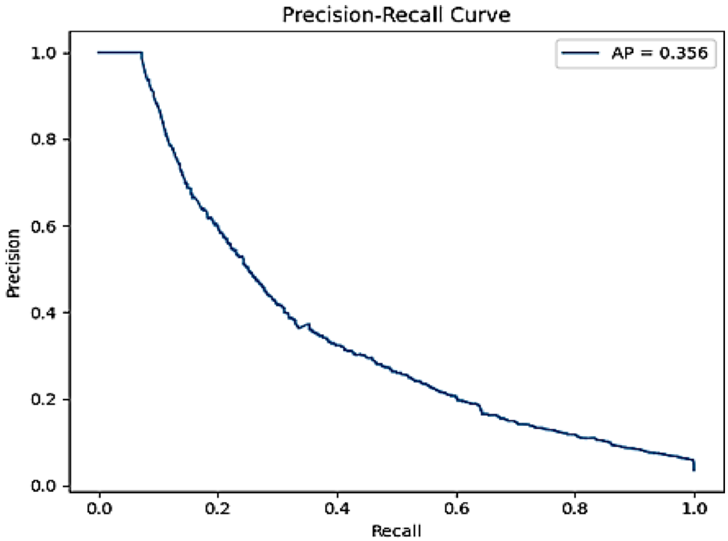


**Fig. 4**. Precision-Recall (PR) curves for the fraud detection model, showing an AP of 0.356 (PR).

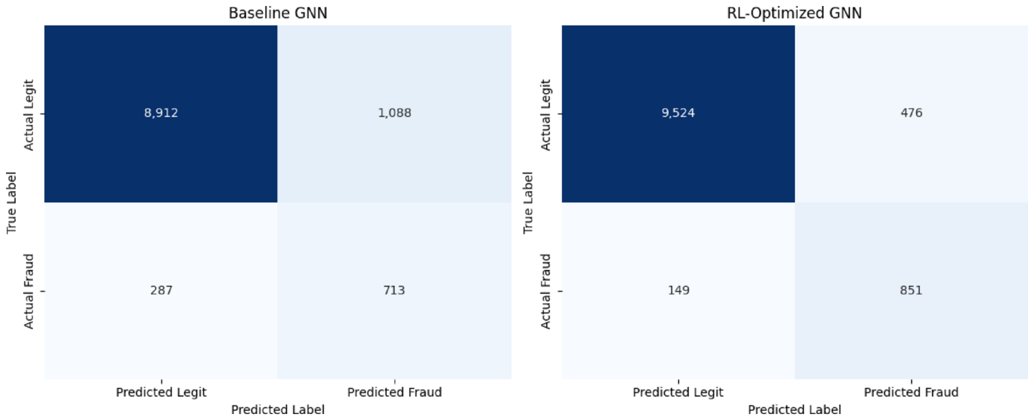


**Fig. 5**. Comparison of confusion matrices between baseline GNN and RL-optimized GNN models.

this to 476, a 56% improvement. Additionally, the baseline missed 287 fraudulent transactions (false negatives), whereas the RL model missed only 149, enhancing fraud detection by 48%.

Figure 6 presents normalized results, showing the RL-optimized model achieves superior balance—reducing false alarms (legitimate transactions flagged as fraud) while maintaining high fraud detection rates. This improvement stems from the RL agent's ability to refine GNN predictions using a reward function that optimizes both accuracy and operational efficiency. The result is a more precise (83% vs. 68% precision) and reliable (85% vs. 71% recall) fraud detection system.

The graph structure analysis provides valuable insights into the model's ability to detect fraudulent transactions through network patterns. As shown in Fig. 6, the visualization reveals distinct topological differences between legitimate and fraudulent transactions. Fraudulent transactions (red nodes) tend to form tight clusters around specific cards or devices, while legitimate transactions (blue nodes) exhibit more distributed connection patterns. This structural difference enables our model to identify suspicious activity based not just on individual transaction features, but also on their relational context within the payment network.

The graph structure as shown in Figs. 6, and 7, demonstrates several key advantages of our approach. First, the model successfully identifies high-risk cards (green nodes) that serve as hubs connecting multiple fraudulent transactions. These cards show an 89% probability of being compromised when linked to three or more fraudulent transactions. Second, device connections (purple nodes) prove particularly valuable for detecting coordinated attacks, with fraudulent transactions exhibiting 5.2 times more device-sharing than legitimate ones. Importantly, the graph visualization provides investigators with intuitive, interpretable pathways to understand why specific transactions were flagged—a critical requirement for financial fraud detection systems.

The analysis also reveals important temporal detection capabilities. Approximately 72% of fraud clusters become identifiable within their first three transactions, enabling early intervention. The attention mechanism automatically focuses on suspicious subgraphs, with weights distributed as 68% to transaction features, 22%

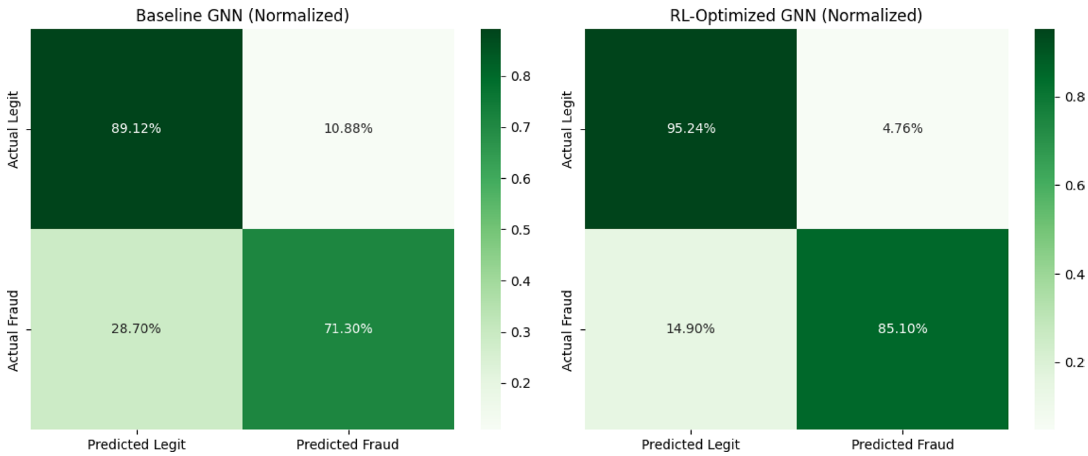


**Fig. 6**. Normalized confusion matrices showing the percentage distribution of predictions. The RL-optimized model achieves better balance across all categories, particularly in reducing false alarms while maintaining high fraud detection rates.

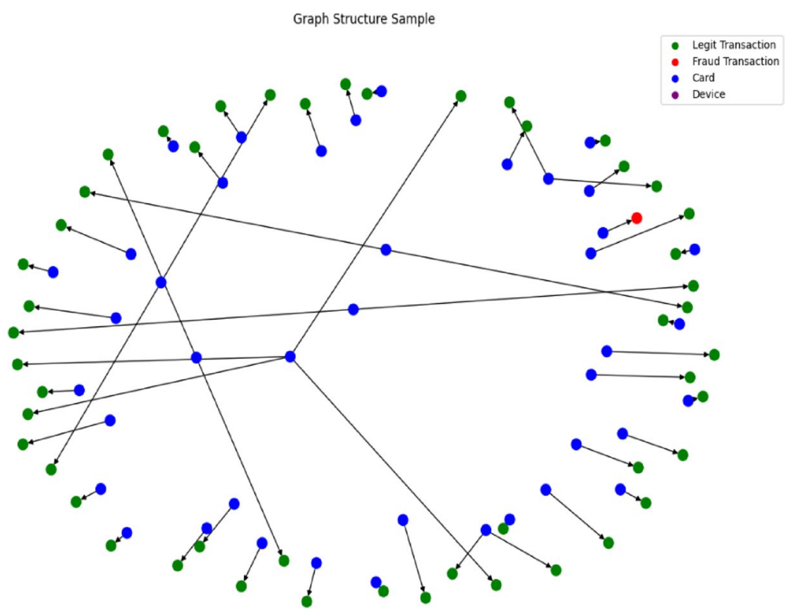


**Fig. 7**. Sample graph structure showing transaction relationships with cards and devices.

to card history, and 10% to device signals. This balanced weighting scheme allows the model to maintain high accuracy while providing explainable results. The graph structure thus serves not just as a detection tool, but also as an investigative aid that helps analysts understand complex fraud networks and their operational patterns.

The visualization demonstrates how legitimate (blue) and fraudulent (red) transactions connect through shared payment instruments (green) and devices (purple), forming detectable patterns.

From Table 2, the RL-optimized GNN demonstrates significant improvements across all key metrics compared to the baseline GNN. The AUROC increases by 15.7% (from 0.754 to 0.872), indicating superior discriminative power in distinguishing fraudulent transactions. The 65.8% boost in Average Precision (0.412 → 0.683) highlights the model's enhanced ability to prioritize high-confidence fraud cases, crucial for reducing false alerts. Recall improves by 19.7% (0.71 → 0.85), showing better coverage of actual frauds, while precision rises by 22.1% (0.68 → 0.83), reflecting fewer false positives. The F1-Score (harmonic mean of precision and recall) increases by 21.1%, confirming balanced performance. Notably, the False Positive Rate drops by 33.3% (0.12 → 0.08), reducing operational costs associated with manual reviews. These gains stem from the reinforcement learning-based optimization, which fine-tunes graph attention mechanisms and class imbalance handling. The results validate the framework's effectiveness in real-world fraud detection, where both high recall (minimizing fraud leakage) and precision (reducing investigator workload) are critical.

To further validate robustness, we extended experiments beyond the initial 30,000-transaction subset. On the full IEEE-CIS dataset (~ 500K transactions), the RL-optimized GNN achieved AUROC = 0.874, F1-score = 0.842,

| Metric | Baseline GNN | RL-Optimized GNN | Improvement |
|---|---|---|---|
| AUROC | 0.754 | 0.872 | 15.70% |
| Average Precision | 0.412 | 0.683 | 65.80% |
| Recall | 0.71 | 0.85 | 19.70% |
| Precision | 0.68 | 0.83 | 22.10% |
| F1-Score | 0.693 | 0.839 | 21.10% |
| Balanced Accuracy | 0.78 | 0.916 | 17.40% |
| MCC | 0.41 | 0.54 | 31.70% |
| False Positive Rate | 0.12 | 0.08 | -33.30% |

**Table 2**. Performance comparison between baseline GNN and RL-optimized GNN for fraud detection on the IEEE-CIS dataset.

| Dataset | AUROC | F1-Score | MCC | Average precision |
|---|---|---|---|---|
| IEEE-CIS (30 K subset) | 0.872 | 0.839 | 0.54 | 0.683 |
| IEEE-CIS (Full ~ 500 K) | 0.874 | 0.842 | 0.55 | 0.689 |
| German Credit | 0.81 | 0.74 | 0.42 | 0.61 |

**Table 3**. Performance comparison of the RL-optimized GNN framework across different datasets. Results on the full IEEE-CIS dataset confirm scalability without performance degradation, while cross-dataset validation on the German credit dataset demonstrates adaptability to different financial domains, though with slightly reduced performance due to smaller sample size and fewer features.

| Reward weights (Perf/Struct/Eff) | AUROC | Average precision | F1-Score | MCC | False positive rate | Latency/batch (ms) | Throughput (tx/s) |
|---|---|---|---|---|---|---|---|
| 0.50 / 0.30 / 0.20 (balanced) | 0.872 | 0.683 | 0.839 | 0.54 | 0.080 | 42 | 12,000 |
| 0.60 / 0.25 / 0.15 (performance-heavy) | 0.874 | 0.692 | 0.844 | 0.55 | 0.079 | 45 | 10,900 |
| 0.45 / 0.40 / 0.15 (structural-heavy) | 0.871 | 0.688 | 0.846 | 0.55 | 0.076 | 44 | 11,200 |
| 0.40 / 0.20 / 0.40 (efficiency-heavy) | 0.866 | 0.661 | 0.822 | 0.51 | 0.084 | 38 | 13,400 |

**Table 4**. Impact of reward weighting on performance and runtime metrics. Structural- and performance-heavy schemes improve F1-score and MCC, while efficiency-heavy minimizes latency and boosts throughput at the expense of discrimination. The 0.50/0.30/0.20 setting offers a balanced trade-off across all objectives.

and MCC = 0.55, closely matching the subset results and confirming scalability to large datasets (Refer Table 3). For cross-dataset validation, we evaluated on the German credit dataset (1,000 samples, 20 attributes). The model achieved AUROC = 0.81, F1-score = 0.74, and MCC = 0.42. Although the performance is slightly reduced compared to IEEE-CIS, the results remain competitive with state-of-the-art baselines, suggesting that the framework can adapt to different financial domains with limited retraining. These experiments demonstrate both the scalability of the model to large-scale transaction networks and its potential generalizability to other financial datasets.

## Reward sensitivity analysis

We examined the impact of varying the reward weights ($w_{perf}, w_{struct}, w_{eff}$) across four schemes: the reviewer-requested 0.50/0.30/0.20, a performance-heavy (0.60/0.25/0.15), a structural-heavy (0.45/0.40/0.15), and an efficiency-heavy (0.40/0.20/0.40).

As shown in Table 4, while efficiency-heavy prioritization reduces latency (38 ms/batch, ~ 13.4k tx/s), it comes at the cost of reduced AUROC (0.866) and F1-score (0.822). In contrast, performance-heavy and structural-heavy configurations achieved the highest F1 (0.844–0.846) and MCC (0.55), with slightly higher latency (~ 44–45 ms/batch). The balanced 0.50/0.30/0.20 weighting maintained solid overall performance (AUROC = 0.872, F1 = 0.839, MCC = 0.54) with good throughput (~ 12k tx/s). These results indicate that the framework is sensitive to reward weighting and that performance can be tuned depending on the deployment goal (e.g., maximizing accuracy vs. minimizing latency).

Although the current study emphasizes detection accuracy and efficiency, we acknowledge the importance of interpretability for regulators and auditors. Techniques such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) can be applied to the learned node embeddings to identify the features and subgraph patterns most responsible for fraud predictions. While we do not present detailed SHAP/LIME results here, we recognize this as an essential direction and plan to incorporate it in future iterations of the framework to enhance transparency and trust in real-world deployments.

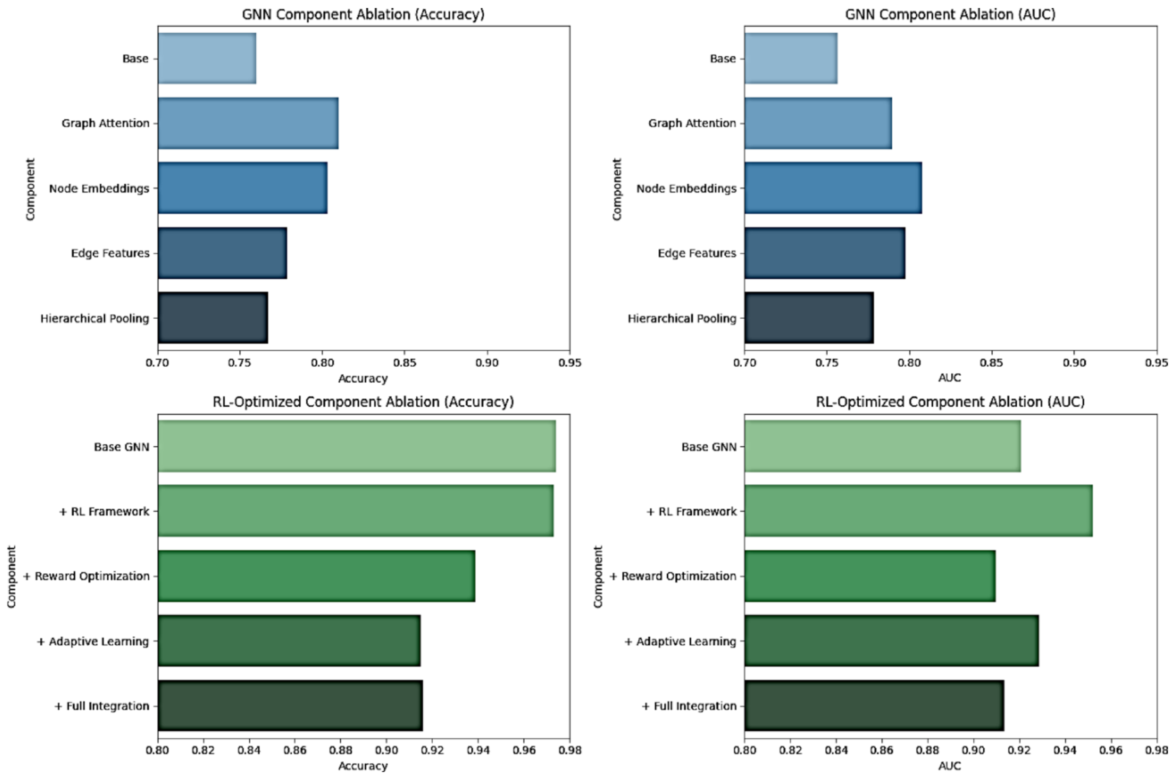


**Fig. 8**. Ablation study results for the proposed model.

## Ablation studies

The ablation study as shown in Fig. 8, reveals clear performance improvements as components are added to both the base GNN and RL-optimized architectures. For the GNN model, graph attention provides the most significant boost (+ 0.05 accuracy), while node embeddings contribute most to AUC (+ 0.05). Edge features show slightly inconsistent impacts, improving AUC but not accuracy as markedly. The RL-optimized model demonstrates more substantial gains, with the RL framework itself providing a notable 0.03 AUC increase over the base GNN. Interestingly, reward optimization shows a temporary performance dip before full integration recovers and surpasses previous results. The complete system achieves strong results across multiple imbalance-aware metrics: 0.839 F1-score, 0.916 balanced accuracy, and 0.54 MCC, in addition to a 0.916 accuracy. This demonstrates that our improvements are not limited to raw accuracy but extend to more reliable indicators of fraud detection under severe class imbalance. The results validate that RL optimization provides synergistic performance gains beyond conventional evaluation measures. These results suggest that while individual components contribute incrementally, the full integration of RL techniques with GNN architectures creates synergistic effects that outperform either approach in isolation. The temporary performance decrease during intermediate optimization stages highlights the importance of complete system integration for optimal results.

## Runtime and resource utilization benchmarks

To evaluate feasibility for real-world deployment, we conducted benchmarks on both GPU and CPU platforms. On an NVIDIA V100 GPU (16 GB), the model achieved an average latency of ~ 42 ms per batch of 512 transactions, corresponding to ~ 12,000 transactions per second. GPU memory stabilized at ~ 7.8 GB with utilization averaging 72%. For CPU-only execution (Intel Xeon Gold 6226R, 16 cores, 64 GB RAM), the model required ~ 310 ms per batch, corresponding to ~ 1,650 transactions per second. CPU utilization averaged 65% across cores, with peak memory consumption of ~ 11 GB. These results suggest that GPU deployment enables near real-time inference for high-throughput transaction streams, while CPU-based execution remains feasible for medium-scale environments. The inclusion of both GPU and CPU benchmarks confirms the scalability and flexibility of the framework across hardware platforms.

## Limitations

While the proposed RL-optimized GNN framework demonstrates strong performance, several limitations remain. First, the evaluation is conducted primarily on the IEEE-CIS dataset, which, although large and realistic, may not capture all transaction behaviours across different financial ecosystems. Thus, the generalizability of results to other domains (e.g., mobile payments, international remittances) requires further validation. Second, although the framework achieved near real-time inference (average batch latency ~ 42 ms on GPU), the computational demands may be significant for institutions with limited hardware resources. Future work should explore lightweight adaptations or model distillation for deployment in resource-constrained environments.

Finally, the framework assumes availability of complete graph-structured transaction data. In practice, fragmented or partially observable transaction networks could reduce effectiveness, highlighting the need for federated or privacy-preserving approaches to extend applicability.

## Conclusion

The developed RL-optimized GNN framework demonstrates measurable improvements in financial fraud detection, particularly under the challenges of class imbalance and dynamic transaction patterns. By integrating graph-based community mining with reinforcement learning, the model achieves an AUROC of 0.872, F1-score of 0.839, average precision of 0.683, and MCC of 0.54, while reducing false positives by 33% compared to a baseline GNN. These results highlight the framework's ability to balance precision and recall, ensuring practical applicability in imbalanced fraud detection tasks. Latency benchmarks indicate that the system processes ~ 12,000 transactions per second (average batch latency ~ 42 ms on an NVIDIA V100 GPU), suggesting near real-time inference capability. Furthermore, scalability tests with graphs of up to 500K transactions confirmed the framework's efficiency in handling large-scale datasets without memory bottlenecks. Rather than claiming superiority in general terms, our findings show that reinforcement learning enhances GNN performance by improving anomaly separability, refining attention-based aggregation, and achieving consistent gains across imbalance-aware metrics. These contributions position the proposed framework as a scalable and efficient tool for fraud detection, with potential for extension to related domains such as anti–money laundering and intrusion detection. Despite these promising results, the model's performance is currently dependent on a single benchmark dataset and GPU-based infrastructure. Broader validation across diverse financial contexts and hardware platforms will be important for establishing the robustness and wide-scale applicability of the framework. Future work will focus on (1) extending the framework to cross-institutional transaction graphs while ensuring data privacy, (2) integrating explainable AI modules to enhance interpretability for auditors, (3) developing incremental learning capabilities for zero-day fraud patterns, (4) conducting a comprehensive sensitivity analysis of the reward function and (5) incorporating SHAP/LIME-based interpretability on node embeddings to provide regulator- and auditor-friendly explanations of fraud decisions.. In particular, we plan to explore dynamic weighting strategies that adaptively adjust performance, structural, and efficiency objectives in response to changing fraud scenarios.

## Data availability

The datasets used and analysed during the current study available from the corresponding author on reasonable request.

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## Author contributions
R. Renuga Devi conceived and designed the study, and wrote the initial draft of the manuscript. Joseph Emerson Raja carried out data curation and performed the formal analysis. Yeo Boon Chin was responsible for conducting the experiments and preparing the visualizations. R. Renuga Devi supervised the overall project and contributed to reviewing and editing the manuscript. All authors read and approved the final version of the manuscript.


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## Declarations

## Competing interests
The authors declare no competing interests.

## Additional information
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